Activity of Vascular Cambium

The vascular cambial ring, when active, cuts off new cells both towards the inner and outer side. The cells which are produced outward form secondary phloem and inward secondary xylem.

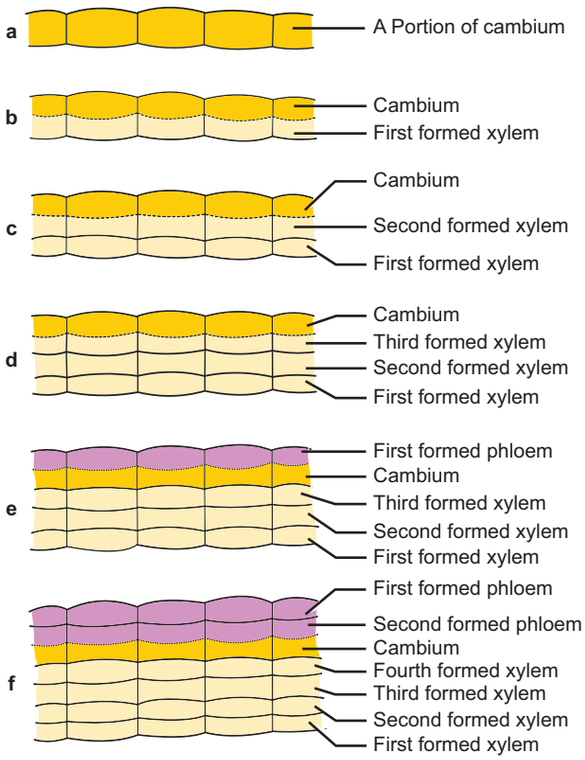


Figure 10.3: Diagrammatic representation of vascular cambial activity (a–f)

At places, cambium forms some narrow horizontal bands of parenchyma which passes through secondary phloem and xylem. These are the rays.

Due to the continued formation of secondary xylem and phloem through vascular cambial activity, both the primary xylem and phloem get gradually crushed.

Secondary Xylem

The secondary xylem, also called **wood**, is formed by a relatively complex meristem, the vascular cambium, consisting of vertically (axial) elongated fusiform initials and horizontally (radially) elongated ray initials.



Xylotomy
The study of wood by preparing sections for microscopic observation.

The axial system consists of vertical files of treachery elements, fibers, and wood parenchyma. Whereas the radial system consists of rows of parenchymatous cells oriented at right angles to the longitudinal axis of xylem elements.

The secondary xylem varies very greatly from species to species with reference to relative distribution of the different cell types, density and other properties. It is of two types.

Porous Wood or Hard Wood

Generally, the dicotyledonous wood, which has vessels is called **porous wood** or **hard wood**. Example: *Morus rubra.*

Non- Porous Wood or Soft Wood

Generally, the gymnosperm wood, which lacks vessels is known as **non- porous wood** or **soft wood**. Example: *Pinus*.

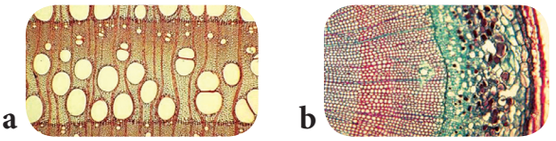


Figure 10.4: Structure of porous (a) and non-porous wood (b)

Differences between Porous Wood and Non-porous Wood

Porous wood or Hard wood, **Example: *Morus***	**Non porous wood or Soft wood,** **Example: *Pinus***
Common in angiosperms	Common in gymnosperms
Porous because it contains vessels	Non-porous because it does not contain vessels

XIth_Botany_Chapter_10_EM.indd 220 5/17/2020 9:17:16 PM